

Write a program to print

1. Fibonacci sequence 0 1 1 2 3 5 8 13 21 ... till nth term

$$2. S = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots + \frac{1}{n}$$

based on user's choice

```
import java.util.Scanner;
class Numbers{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);

        System.out.println("Enter n");
        int n= sc.nextInt();

        System.out.println("Enter 1 for Fibonacci sequence and 2 for sum of series");
        int choice = sc.nextInt();

        switch(choice){
            case 1:
                int n1=0, n2=1;
                System.out.print(n1+ " " +n2);
                for(int i=1; i<=n; i++){
                    int s=n1+n2;
                    System.out.print(s);
                    n1=n2;
                    n2=s;
                }
                break;

            case 2:
                double s=0;
                for(int i=1; i<=n; i++){
                    s+= 1/i;
                }
                System.out.println("Sum of Series= "+s);
                break;

            default: System.out.println("Wrong Input");
        }
    }
}
```

Value Description Table

Data type	Variable	Description
int	n	to store the nth term
int	choice	to store choice given by user
double	s	to accumulate the sum of series
int	i	to iterate upto n
int	n1	to store first number of Fibonacci
int	n2	to store second number of Fibonacci